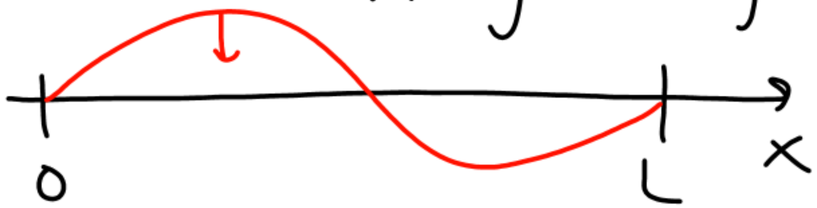
Wave Equation

1D case $u(x,t)$

$u(x,t)$ = vertical displacement of vibrating string



$$\begin{cases} u_{tt} - u_{xx} = 0 \\ u(x,0) = g(x) \\ u_t(x,0) = h(x) \end{cases}$$

Also need boundary conditions

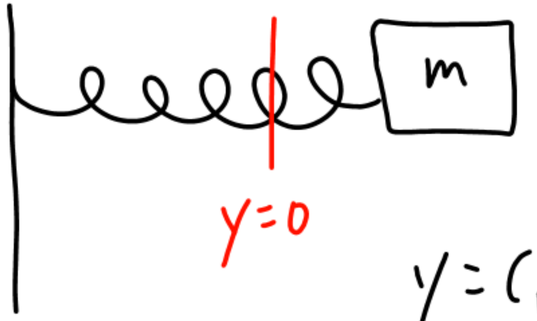
if PDE is posed on bounded domain

like $u(0,t) = 0$ (Dirichlet)

$u(L,t) = 0$

(Harmonic oscillator:

ODE $y'' = -\frac{k}{m} y$, $y(0) = a$, $y'(0) = b$



$$y = c_1 \cos\left(\sqrt{\frac{k}{m}}\, t\right) + c_2 \sin\left(\sqrt{\frac{k}{m}}\, t\right)$$

)

General dimension $n \geq 1$:

$$\begin{cases} u_{tt} - \Delta u = 0 \\ u(x,0) = g(x) \\ u_t(x,0) = h(x) \end{cases} \qquad x \in \mathbb{R}^n,\ t > 0$$

Today: Solution formulas for: $n=1$
$n=3$
$n=2$

$n=1$: $\begin{cases} u_{tt} - u_{xx} = 0, & x \in \mathbb{R},\ t > 0 \\ u(x,0) = g(x) \\ u_t(x,0) = h(x) \end{cases}$

Claim: $u_{tt} - u_{xx} = \left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x}\right)u$

Why? $\left(a^2 - b^2 = (a-b)(a+b)\right)$ $\left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x}\right)\left(u_t - u_x\right) = u_{tt} - u_{xt} + u_{tx} - u_{xx}$
$= u_{tt} - u_{xx}$.

$$0 = u_{tt} - u_{xx} = \left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x}\right)\underbrace{\left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x}\right)u}$$

Let $\underline{v = u_t - u_x} = \frac{\partial}{\partial t}u - \frac{\partial}{\partial x}u$

$\longrightarrow \left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x}\right)v = 0$, i.e. $v_t + v_x = 0$

$z(s) = v(x+s, t+s)$

$\frac{d}{ds}z = 0 \longrightarrow z(0) = z(-t)$

$v(x,t) = v(x-t, 0)$

What is $v(x,0)$?

$$v(x,0) = u_t(x,0) - u_x(x,0)$$
$$= h(x) - \frac{\partial}{\partial x}g(x)$$
$$= h(x) - g'(x)$$

$$\longrightarrow \boxed{v(x,t) = h(x-t) - g'(x-t)}$$

$$v(x,t) = h(x-t) - g'(x-t)$$

$u_t - u_x = v$ is another transport equation.

$$z(s) = u(x-s, t+s)$$

$$\frac{d}{ds} z = -u_x + u_t = v(x-s, t+s)$$

$$z(0) - z(-t) = \int_{-t}^{0} v(x-s, t+s)\, ds$$

$$u(x,t) - u(x+t, 0) = \int_{-t}^{0} \left[h(x-s-(t+s)) - g'(x-s-(t+s)) \right] ds$$

$$u(x,t) = g(x+t) + \int_{-t}^{0} \left[h(x-t-2s) - g'(x-t-2s) \right] ds$$

$$u(x,t) = g(x+t) + \int_{-t}^{0} \left[h(x-t-2s) - g'(x-t-2s) \right] ds$$

$$= g(x+t) + \int_{-t}^{0} h(x-t-2s)\,ds - \left. \frac{g(x-t-2s)}{-2} \right|_{s=-t}^{s=0}$$

$$= g(x+t) + \frac{1}{2} g(x-t) - \frac{1}{2} g(x+t) + \int_{-t}^{0} h(x-t-2s)\,ds$$

$$= \frac{1}{2}\left(g(x+t) + g(x-t) \right) + \int_{x+t}^{x-t} h(y) \frac{dy}{-2}$$

$y = x - t - 2s$

$dy = -2\,ds$

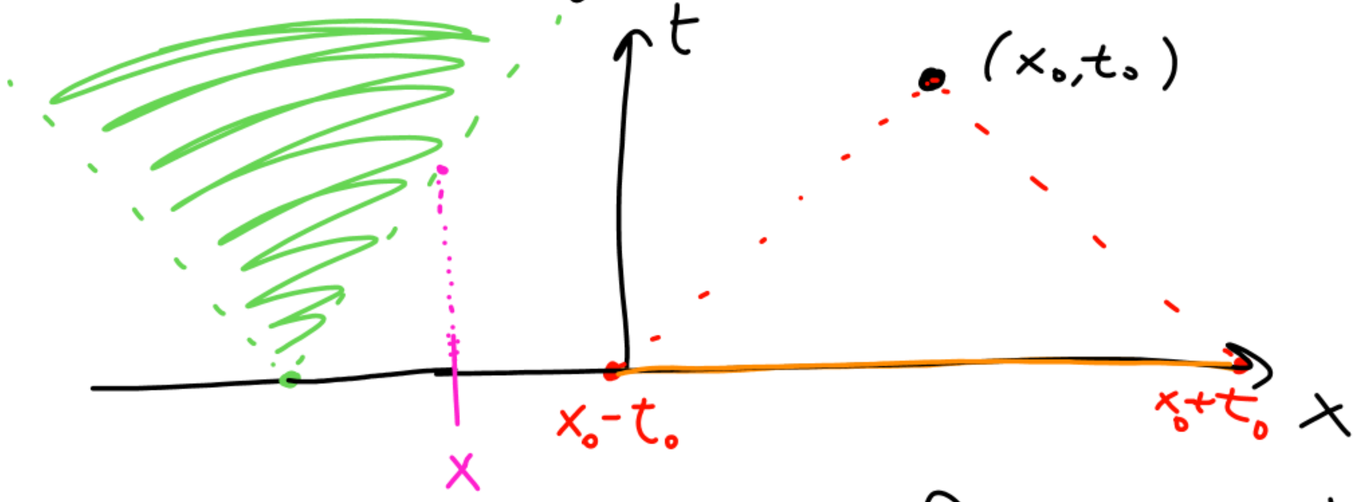
$s = 0 \rightarrow y = x - t$

$s = -t \rightarrow y = x + t$

$$\boxed{u(x,t) = \frac{1}{2}\left(g(x+t) + g(x-t) \right) + \frac{1}{2} \int_{x-t}^{x+t} h(y)\,dy}$$

d'Alembert's Formula

$$u(x,t) = \frac{1}{2}\left(g(x+t) + g(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy$$



This finite speed of propagation is different from the heat equation.

$$\left(u(x,t) = \sum a_n(t)\sin(\lambda_n t). \right)$$

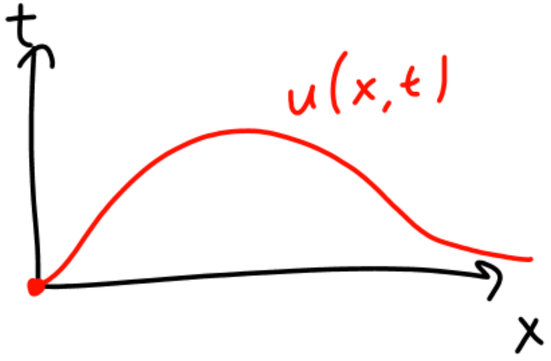
$$u_{tt} - c^2 u_{xx} = 0$$

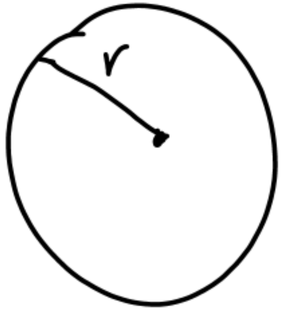
General wave speed.

The regularity of u is limited by the regularity of g & h

(Unlike the heat equation, which regularizes solutions.)

Solution for $x \in [0, \infty)$ via <u>odd reflection</u>:

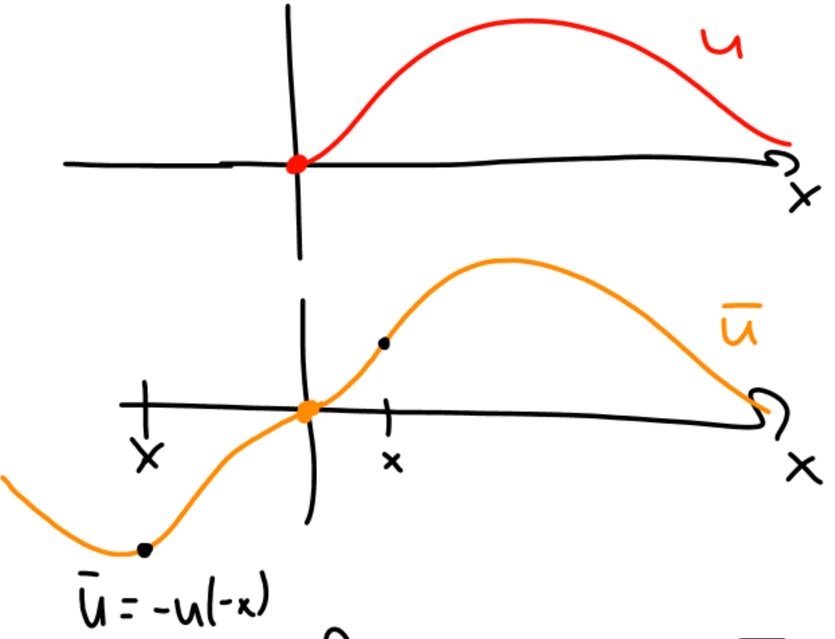




$$\begin{cases} u_{tt} - u_{xx} = 0 \quad , \quad x > 0, \ t > 0 \\ u(x,0) = g(x) \\ u_t(x,0) = h(x) \\ u(0,t) = 0 \end{cases}$$

Assume $g(0) = h(0) = 0$.

If u is a solution, consider $\bar{u}(x) = \begin{cases} u(x,t), & x \geq 0 \\ -u(-x,t), & x < 0 \end{cases}$



Claim: $\bar{u}$ solves

$$0 = \bar{u}_{tt} - \bar{u}_{xx} \quad \text{in } \{x \in \mathbb{R},\ t > 0\}$$

with $\bar{u}(x,0) = \bar{g}(x)$, $\bar{u}_t(x,0) = \bar{h}(x)$ → odd reflections of g, h

If $x > 0$, $\bar{u}_{tt} - \bar{u}_{xx} = u_{tt} - u_{xx} = 0$ ✓

If $x < 0$, $\bar{u}_{tt} - \bar{u}_{xx} = \frac{\partial^2}{\partial t^2}\left(-u(-x,t)\right) - \frac{\partial^2}{\partial x^2}\left(-u(-x,t)\right)$

$$= -u_{tt}(-x,t) + u_{xx}(-x,t) = 0 \quad ✓$$

At $x=0$, $\bar{u}_{tt}=0$ since $\bar{u}(0,t)=0$ for all t.

$\bar{u}_{xx}=0$ because graph must have a point of inflection.

$\Rightarrow$ $\bar{u}$ solves Wave Eqn on $\mathbb{R}$

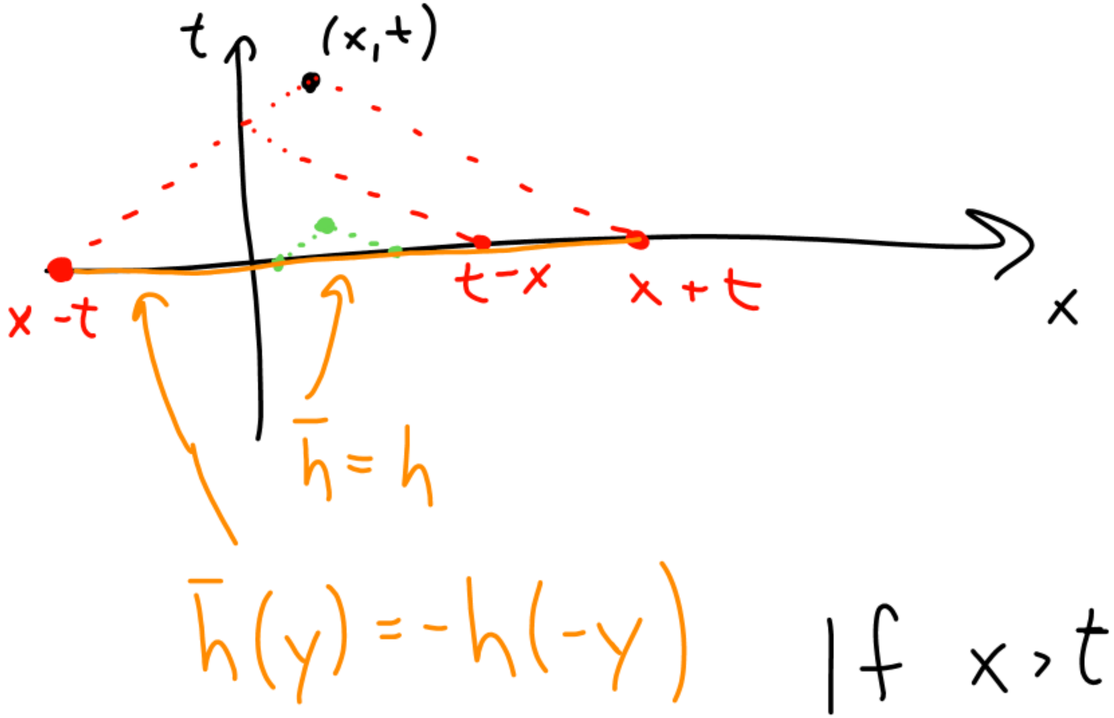
d'Alembert: $\bar{u}(x,t)=\frac{1}{2}\left(\bar{g}(x+t)+\bar{g}(x-t)\right)+\frac{1}{2}\int_{x-t}^{x+t}\bar{h}(y)\,dy$

For $x>0$, $u=\bar{u}$

$\Rightarrow u(x,t)=\frac{1}{2}\left(\bar{g}(x+t)+\bar{g}(x-t)\right)+\frac{1}{2}\int_{x-t}^{x+t}\bar{h}(y)\,dy$

$$u(x,t) = \frac{1}{2}\left(\bar{g}(x+t) + \bar{g}(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} \bar{h}(y)\,dy$$

$x > 0,\ t > 0,$ so $x+t > 0,$ so $\bar{g}(x+t) = g(x+t)$.



If $x > t$, then $x - t > 0$,
so $\bar{g}(x-t) = g(x-t)$

If $x < t$, then $x - t < 0$,
so $\bar{g}(x-t) = -g(t-x)$

If $x > t$, $\displaystyle\int_{x-t}^{x+t} \bar{h}(y)\,dy = \int_{x-t}^{x+t} h(y)\,dy$

If $x < t$,

$$\int_{x-t}^{x+t} \bar{h}(y)\,dy = \int_{x-t}^{0} \bar{h}(y)\,dy + \int_{0}^{x+t} \bar{h}(y)\,dy$$

$\bar{h}(y) = -h(-y)$ $\qquad$ $\bar{h}(y) = h(y)$

$$= -\int_{x-t}^{0} h(-y)\,dy + \int_{0}^{x+t} h(y)\,dy$$

$z = -y$

$dz = -dy$

$$= \int_{t-x}^{0} h(z)\,dz + \int_{0}^{x+t} h(y)\,dy = -\int_{0}^{t-x} h(z)\,dz + \int_{0}^{x+t} h(y)\,dy$$

$$= \int_{t-x}^{t+x} h(y)\,dy$$

Solution:

$$u(x,t) = \begin{cases} \frac{1}{2}\left(g(x+t)+g(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy, & x \geq t, \\ \frac{1}{2}\left(g(x+t)-g(t-x)\right) + \frac{1}{2}\int_{t-x}^{t+x} h(y)\,dy, & 0 \leq x < t. \end{cases}$$

Colloquium Talk: 4pm Today, CRF 210